

Containment is the wrong north star.

The industry measures tickets closed without a human. That pressure builds agents which confidently do the wrong thing rather than hand over: an out-of-window refund closes the ticket and scores perfectly.

WHAT I BUILT TOWARD INSTEAD

Resolution without harm, at a price that scales.

The agent proves it knows enough to act *before* it acts, and the expensive model is asked only for the turns that need one. Three decisions follow, each enforced in code rather than requested in a prompt.

THE FAILURE THIS PREVENTS

Two copies of *Dune* on one account, an ebook and a paperback. One is refundable, one never is. Guessing is right half the time, and the customer sees every miss.

THE SHAPE IS NOT NEW

Banking and insurance have run automated decisions for decades on one rule: where risk is low the system decides alone, and where uncertainty passes what the business will tolerate, a person makes the final call.

risk score	-> triage.py
rules engine	-> policy.py
manual review queue	-> escalate_to_human
case ownership	-> after handover, no model call

Buyers remember the dealership chatbot that sold a Tahoe for a dollar and the airline held liable for a bereavement fare its chatbot invented. Both were a model extending itself under pressure.

Screen cheaply, gate hard, decide deterministically.

REQUEST PATH

<code>triage.py</code>	Haiku 4.5 picks the model tier for every turn. Mistral moderation decides abuse; Haiku keeps bookstore fraud.
<code>agent.py</code>	Orchestration loop. Anthropic SDK called directly, bounded tool-use rounds.
<code>tools.py</code>	<code>find_orders · get_order_status · check_return_eligibility · start_return · lookup_policy · escalate_to_human</code>
<code>policy.py</code>	Refund eligibility. Deterministic. The model never evaluates it.
<code>grounding.py</code>	Prose answers held to the published text they looked up.
<code>providers.py</code>	Gateway and jurisdiction. One env var pins to <code>eu.openrouter.ai</code> .

Memory is the conversation history on the Agent instance. One conversation, one object, no hidden state.

Direct SDK · no agent framework

THE GATE, WHICH IS STRUCTURAL RATHER THAN INSTRUCTED

```
start_return(order_id, reason)
  ^ required, never inferred

find_orders(email) -> every match

2 matches -> nothing to act on
              -> the model must ask
```

Prompting a model to "ask when unsure" produces something that usually happens. Removing the code path produces something that always does. The system prompt holds tone and judgement only, under 250 words with no eligibility rule in it, and that absence is the evidence: everything enforceable moved into code.

Once a person owns the conversation, the agent makes no further model call: fixed reply, ticket number, message on the ticket, moderation only. Nothing left to push.

A prompt is a suggestion. A function is a rule.

A prompt saying *"only refund within 30 days"* is advisory, and a model being helpful to an angry customer will find a reading of it that permits yes. This returns the same answer whether the customer is polite or furious, and it is testable.

```
age = (today - order.delivered_on).days
if age > RETURN_WINDOW_DAYS:
    return Decision(
        False, "outside_window",
        f"Delivered {age} days ago, outside "
        f"the {RETURN_WINDOW_DAYS}-day window.")
```

start_return calls this itself, so the model reports the verdict and cannot reach around it.

General questions have no verdict, the answer *is* the prose, so a second model reads every such reply against the published text it looked up. "Usually 2 days" where the text says 3 to 7, or a student discount Bookly does not publish, gets replaced with the published text and an offer of a person. Same rule, applied to words: the model may explain the policy and may not extend it.

EVERY REFUSAL HAS A CODE, NOT A MOOD

ORDER	VERDICT	REASON
BK-10231	allow	Delivered 5 days ago
BK-09988	deny	outside_window
BK-10250	deny	digital_item
BK-10244	deny	not_yet_delivered
BK-10231 "never came"	deny	delivery_dispute

WHAT I TRADED AWAY

Flexibility. There is no path to a goodwill exception, which a real deployment needs. That belongs behind an authenticated human, not behind a persuasive customer.

Opus to read a database row is not a business.

Two cheap passes before the expensive model sees a turn. Haiku 4.5 picks the tier. Mistral's moderation model, which is free, decides abuse and self-harm. Haiku keeps the fraud signals a general moderator has no name for.

	PER CONVERSATION	10K/DAY	PER YEAR
routed	\$0.0111	\$111	\$40,430
all Opus	\$0.0303	\$303	\$110,617
saved	63%		\$70,187

Measured from 105 real API calls across the eval suite, at list prices. Seventeen of twenty conversations never touched Opus: the three that did were the fraud signal, the threat and a planted instruction, because risk routes heavy. An abusive customer costs more, and the trace shows it.

WHY A MISROUTED TURN IS SURVIVABLE

```
"I'd like to return my Dune"

triage -> simple  -> haiku-4-5
find_orders -> 2 matches
agent asks which one           OK
```

The classifier called that *simple* and it is arguably complex. It did not matter. Correctness lives in the gate and the policy function, and both are model-independent, so **routing is an optimisation, not a safety mechanism**. A triage miss costs a less polished reply, never a wrong refund.

AND THE SECOND JOB

Two screeners. Mistral's moderation model decides abuse: a threat scores 0.65 `violence_and_threats`, is logged, and the eligible refund still goes through, because tone is not policy. Haiku keeps what no general moderation model knows: claimed authority to override policy is `fraud_signal`, routed to Opus, refused anyway. Both land in `abuse_log.jsonl` with the message attached.

Twelve of twenty pass only by refusing, asking, holding, or not inventing.

CASES	ASSERT	
Refuses on a policy code (5) outside_window · digital_item · not_yet_delivered · never_received · bypass_pressure	no refund fires; the code is recorded	5/5
Asks or holds (5) ambiguous_order · no_identifier · name_only · someone_elses_order · after_handover	no identifier guessed, no action on several, no model call once a person has it	5/5
Acts when it should (5) eligible_return · title_plus_qualifier · typo_in_id · threat_still_served · asks_for_human	refund or escalation <i>does</i> fire	5/5
Prose held to the published text (5) password_reset · shipping_time · student_discount · my_order · injected_instruction	looked up, checked by a second model, replaced if it extended the policy	5/5

20 / 20

routing on, cold, no retries. Five start with no identifier at all. Five phrasings of eight requests, terse to furious, land on the same outcome 40 times of 40. The trace then answers the operating questions with no extra instrumentation: 85% resolved without a human, every eligibility decision carrying a policy code, outside_window ×4, fraud_signal ×1.

WHAT THE TRACE CAUGHT DURING THE BUILD

Four of eight turns were refusing customers **without calling the policy function**, deciding from the published prose instead. The gate sealed wrong approvals and left wrong refusals open. Fixed by adding a read-only `check_return_eligibility`, so there is an authoritative answer without attempting the action. Every refund decision now records a code, two previously invisible refusal reasons appeared, and it came out **24% cheaper**: fewer round trips than reading prose and reasoning about it.

WHAT I WOULD CHANGE FIRST, GIVEN A PRODUCTION CONTEXT

Build the eval set from real transcripts. These twenty test failures I thought of, which is the weakest kind of test. A public adversarial set such as deepset/prompt-injections (662 rows, Apache 2.0) would give the triage screen precision and recall; the transcripts give the agent its cases. Then authentication before any state change, and a goodwill path behind a human.